TRAINEX | ESP32 Hardware Wiring Guide

Hardware scope: seven normally-open momentary switches and one AS5600 13-notch master handle. Only these controls are wired to the ESP32. This is a prototype control interface, not a railway safety system.

1. ESP32 pin assignments

Control	ESP32 pin	Connection
Left Door Enable	GPIO18	N.O. button -> GND
Left Door Open	GPIO26	N.O. button -> GND
Left Door Close	GPIO27	N.O. button -> GND
Right Door Enable	GPIO19	N.O. button -> GND
Right Door Open	GPIO25	N.O. button -> GND
Right Door Close	GPIO33	N.O. button -> GND
Vigilance Acknowledge	GPIO23	N.O. button -> GND
AS5600 SDA / SCL	GPIO21 / GPIO22	I2C to sensor SDA / SCL
AS5600 Power	3V3 + GND	3.3 V sensor supply, common ground

2. Breadboard assembly

1. Unplug USB-C. Put the ESP32 beside the breadboard, then connect an ESP32 GND pin to the blue negative rail.

2. If the blue rail is split in the center, bridge the two segments with a jumper. Connect other ground rails only if used. Test continuity.

3. Connect ESP32 3V3 to the red power rail for the AS5600 only. Never join red (+) and blue (-) rails; never put 5 V on the ESP32 pins.

4. Wire sensor: VCC to 3V3, GND to GND, SDA to GPIO21, SCL to GPIO22. Verify breakout I2C pull-ups are at 3.3 V.

5. Mount all seven normally-open switches. One terminal connects to its assigned GPIO; the other connects to the common GND rail. For four-leg tactile switches, use contacts that are open until pressed.

3. Website command mapping

Physical control	WSS controlId	Value
Left Enable	DOOR_ENABLE_LEFT	true
Left Open	DOORS_OPEN	"LEFT"
Left Close	DOORS_CLOSE	"LEFT"
Right Enable	DOOR_ENABLE_RIGHT	true
Right Open	DOORS_OPEN	"RIGHT"
Right Close	DOORS_CLOSE	"RIGHT"
Vigilance	VIGILANCE_ACKNOWLEDGE	true

Physical control	WSS controlId	Value
Master handle	MASTER_HANDLE	"EB" / "B7" ... "P4"

Door OPEN/CLOSE uses LEFT or RIGHT strings. The complete project includes the necessary browser and relay validation patches; earlier versions rejected these string values.

TRAINEX | Handle Calibration and Verification

4. Calibrate the AS5600 master handle

1. With the handle at Neutral (N), power the ESP32 by USB-C, open PlatformIO Serial Monitor at 115200 baud, and locate AS5600 RAW.

2. Copy the 0-4095 reading into TRAINEX_AS5600_NEUTRAL_RAW in include/TrainexHardwareConfig.h. By default it is -1 and intentionally disables commands.

3. Rebuild/upload. Move to P1. If the direction is reversed, set TRAINEX_AS5600_REVERSE to 1, then rebuild.

4. Check all 13 positions. The physical angles are relative to Neutral and spaced every 12 degrees.

5. Notch schedule

Power/Neutral	Degrees	Brake/EB	Degrees
P4	+48	B1	-12
P3	+36	B2	-24
P2	+24	B3	-36
P1	+12	B4	-48
N	0	B5	-60
-	-	B6	-72
-	-	B7	-84
-	-	EB	-96

6. Testing and website pairing

Power on only after checking GND continuity, power-rail separation, and the absence of permanent shorts across each pushbutton.

Press each button once and inspect USB Serial Monitor logs: LEFT/RIGHT enable; open/close independently; vigilance acknowledgement. Release does not send another command.

Start a TRAINEX simulation session from the Hardware tab, authenticate and pair the ESP32 over the deployed 2.4 GHz Wi-Fi + WSS relay. Web engine enforces standstill, Neutral and door-enable safety rules.

Monitor the TMS door graphics: LEFT doors above the train illustration; RIGHT doors below. Indicator lamps and gauges are virtual only.

7. Excluded physical hardware

Do not connect old GPIO25 EB release, GPIO33 horn, GPIO32 buzzer, or GPIO13 security-brake circuits. There are no physical lamp outputs. EB remains the EB notch on the master handle; EB release is handled through the existing website interface.

The materials reference includes fewer distinct pushbutton functions than this seven-button scope. Check actual button inventory and obtain additional N.O. assemblies if required. The AS5600 also needs the correct diametrically magnetized shaft-end magnet.